

Data Engineering Project

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Content

Introduction Project End To End

Project Overview

Project ini merupakan implementasi end-to-end data engineering pipeline untuk mengolah data e-commerce Brazil dari Olist



Tujuan Project

- Membangun data pipeline yang scalable dan reliable
- Mengolah data transaksi e-commerce secara efisien
- Menyediakan data warehouse untuk analisis bisnis
- Implementasi best practices dalam data engineering

Real-World Application: Pipeline ini mensimulasikan proses data engineering di industri e-commerce yang sesungguhnya



olist

Dataset: Olist Brazilian E-Commerce

Dataset Components

Orders: 100k+ transaksi pelanggan

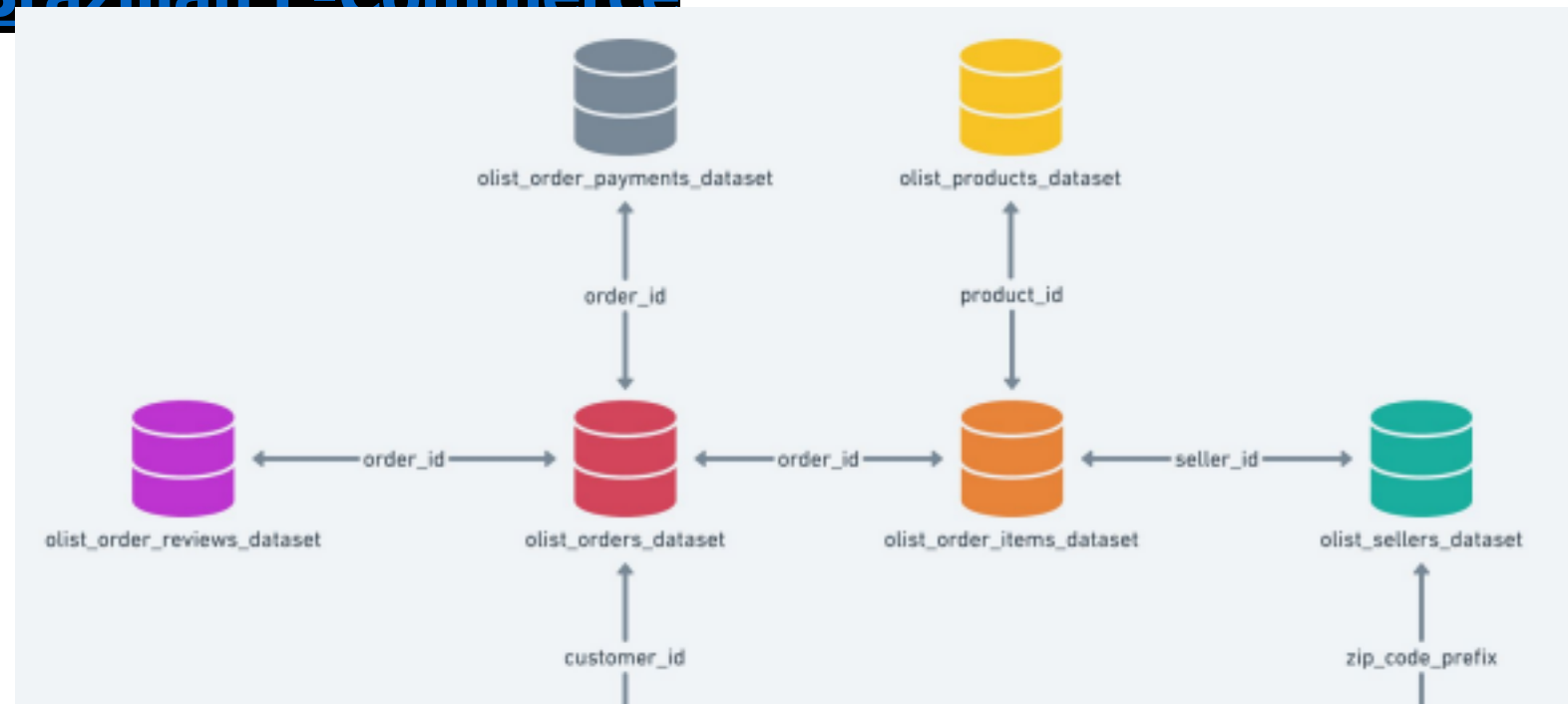
Order Items: Detail produk dalam setiap order

Customers: Informasi pelanggan

Products: Katalog produk

Sellers: Data penjual

Payments: Metode dan detail pembayaran



Reviews: Rating dan review pelanggan

Geolocation: Data geografis Brazil



Python: Fleksibel untuk data extraction dan automation

PySpark: Handle large-scale data processing dengan distributed computing

Airflow : Batch scheduler

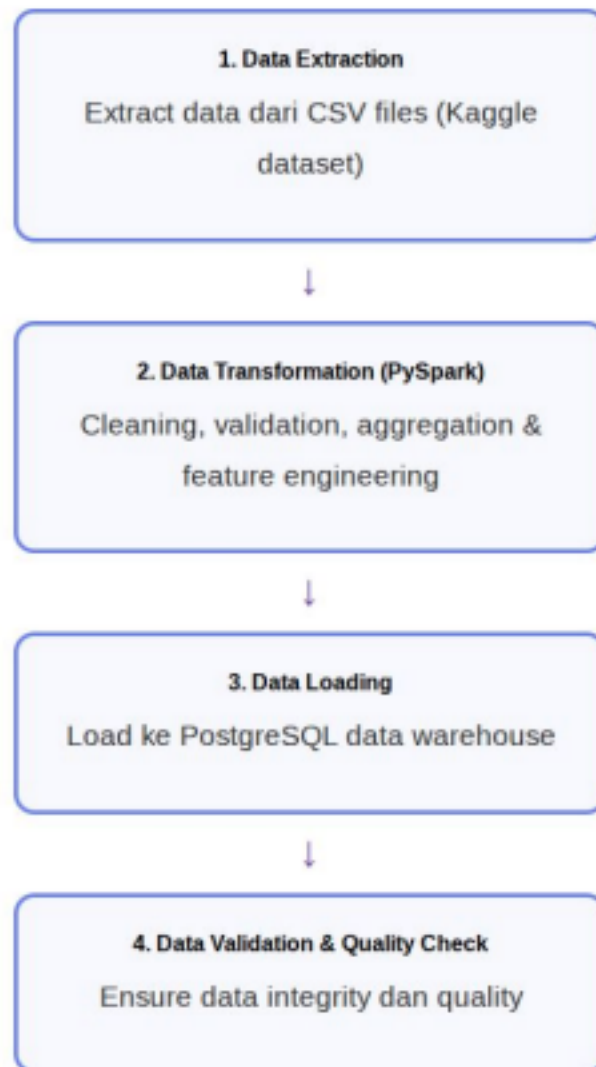
Kafka : Real Time Streaming Data

PostgreSQL: Reliable relational database untuk analytical queries

End to End Pipeline



Pipeline Stages



Expected Result

Fact Tables

Sales transactions dengan semua dimensi terkait

Dimension Tables

Customer, Product, Seller, Time dimensions

Analytics Views

Pre-aggregated data untuk quick insights

KPI Metrics

Revenue, conversion rate, customer lifetime value

Geospatial Analysis

Sales distribution by region

Product Insights

Top products, categories, review analysis

Project Deliverables

Technical Deliverables

ETL Pipeline Code: Modular Python & PySpark scripts

Database Schema: Optimized star schema design

Data Quality Framework: Validation & monitoring

Scheduler : Create scheduler to ingest data to database

Documentation: Architecture dan setup guide

Business Deliverables

Sales Dashboard Data: Ready untuk visualisasi

Customer Segmentation: RFM analysis

Seller Performance Metrics: Rating & sales analysis

Product Recommendation Data: Based on purchase patterns

Success Metrics: Pipeline dapat memproses 100k+ records dengan execution time < 10 menit

Pengumpulan



1. github : Berisikan code yang terstruktur

2. DB Diagram component



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